**Advanced Business Analytics**
K62 Digital business

GROUP 1

# VIETNAM REAL ESTATE AVM

An independent estimate of fair value for every resale home in Hanoi and HCMC

**60%–64%**

Apartment and alley-house estimates within ±20% of asking price

**1.5×**

Top 10% of risk scores:
stale rate vs the test-set average
(unsold for over 180 days)

**VND 9.7 bn**

Net partner value per year
Base-case estimate

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# THE PROBLEM

**Homes cost billions of VND, but talks start from the seller's asking price.**

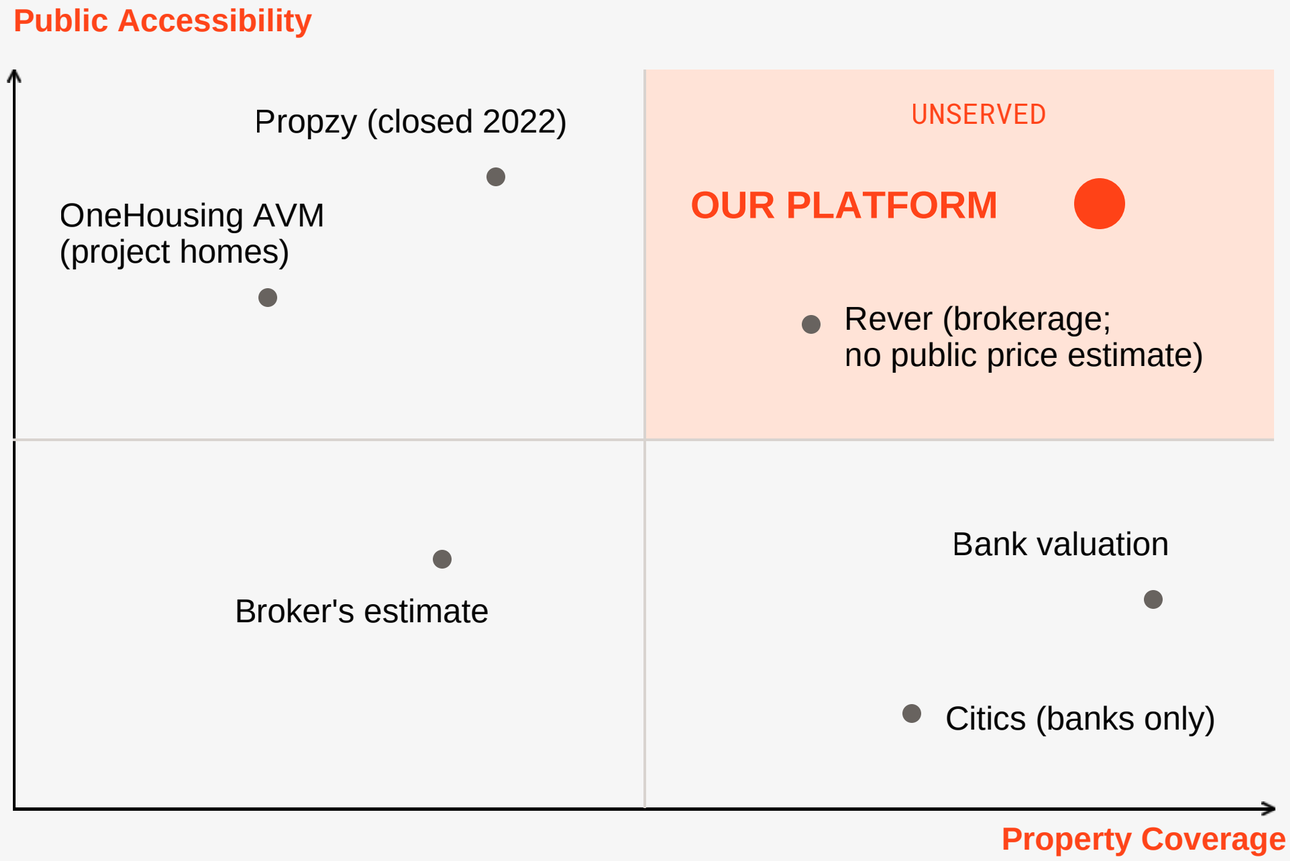
Brokers compare similar homes by hand. Banks wait for manual valuations.

The missing price guide hurts four teams in different ways.




Source: team analysis.

# MARKET POSITIONING




## OneHousing

533,856 homes in 627 projects one month after the 2022 launch. Reports error below 5%. Techcombank partner.

## Citics

More than 32 banks use it. Raised USD 2.1 million in 2025. For institutions only.

Rever is a broker, with no public valuation. Propzy closed in 2022.

**Public values for all property types. Method and error by type are published. Uses listings and map data.**

Source: company websites and media reports; claims are not independently verified.

One table supports three decisions.

# THE SYSTEM

## P

### Problem

Buyers, brokers and lenders in Hanoi and HCMC have no independent price guide for resale homes.

## D

### Data

5,268 mogi.vn sale listings linked to 34,321 OpenStreetMap points: metro, schools, parks, markets and city centre.

## M

### Method

**Fair Value Engine**
Regression + gradient boosting
**Stale Listing Radar**
Unsold-risk model
**Market Map**
District groups

## T

### Tool

Reproducible local workflow and browser-based dashboards.
Planned: cloud database, scheduling, monitoring, and model approval.

## DECISIONS

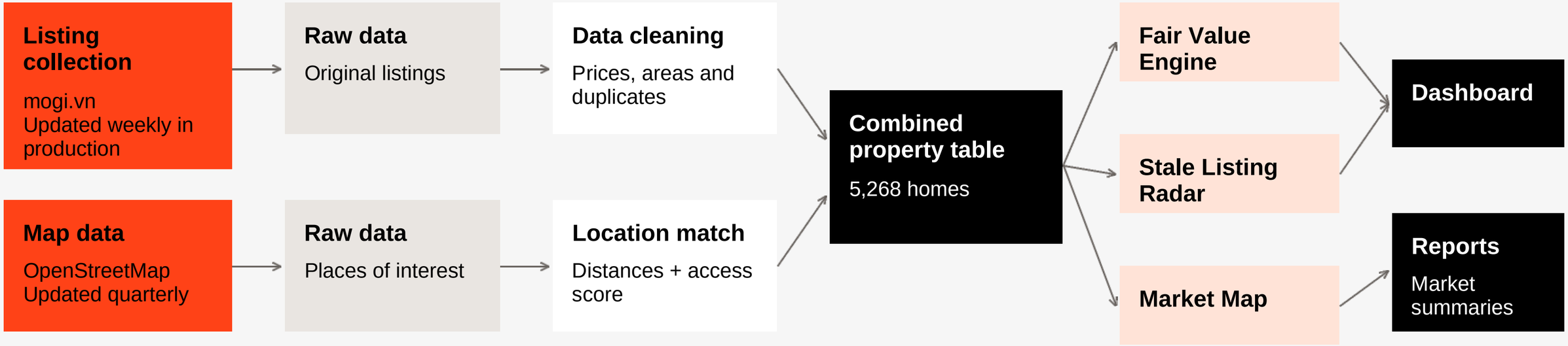
**Fair value**
**+ 90% range**
**Unsold-risk rank**
**District group**

Price, advise sellers, lend and allocate

Source: mogi.vn listings, 15 Oct 2026 (N = 5,268).

5,268 homes linked to 34,321 map points.

# DATA PIPELINE




## 6,000 listings in 46 minutes. 88% pass data checks.

Prototype collection window: 1.5 hours. Speed: 7,792 listings/hour, with no blocks. At least 33,000 listings take about 5 hours.

Prototype: laptop files. Production: cloud database with location queries, scheduled runs and model versions with accuracy records.

Source: mogi.vn listings, 15 Oct 2026 (N = 5,268); OpenStreetMap contributors (ODbL).

Three linked engines support three decisions.

# MODEL ARCHITECTURE

**Combined property data**

Size, rooms, legal status, distances and district

## Fair Value Engine

Regression + boosted trees; a 90% price range

→ **Set prices; screen loans**

Fair value + range; first collateral check

↓

## Stale Listing Radar

Predicts risk of staying unsold over 180 days

→ **Prioritise; advise sellers**

Weekly list of high-risk listings

↓

## Market Map

Groups districts by their market

→ **Target areas; assign brokers**

Where to advertise and add brokers

All accuracy scores use homes unseen in training. Age starts when a listing is created: sellers change posted dates, and 59% re-posted after 30 days.

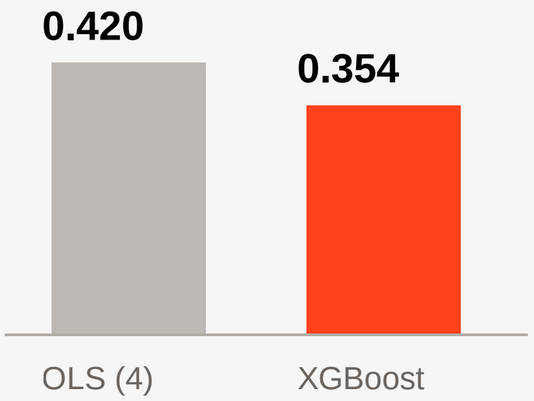
53.9% of 1,054 unseen homes are valued within ±20% of asking price.

# FAIR VALUE ENGINE

XGBoost beats regression on all four measures

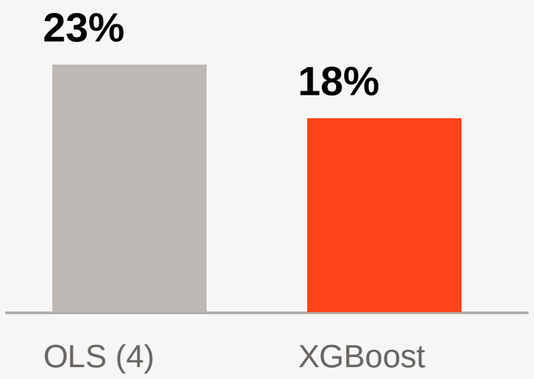
### Log-price error (RMSE)

LOWER IS BETTER



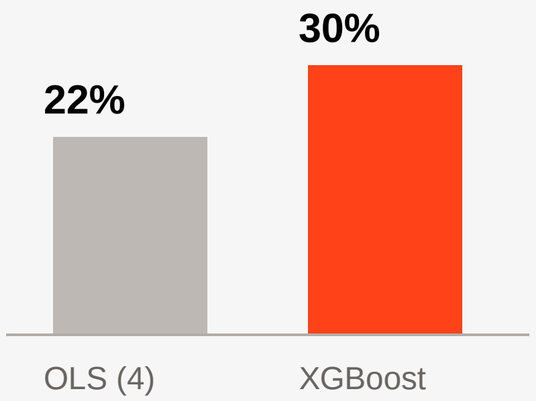

### Median absolute % error

LOWER IS BETTER



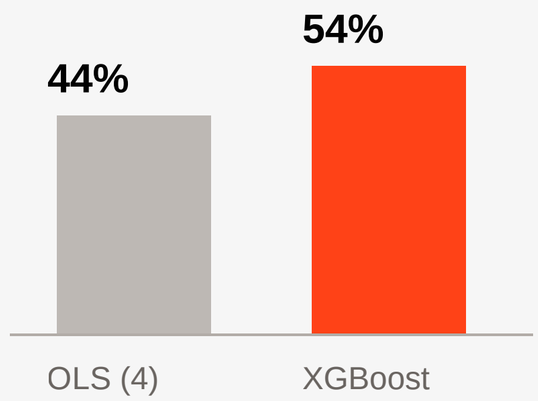

### Within ±10% (PPE10)

HIGHER IS BETTER




### Within ±20% (PPE20)

HIGHER IS BETTER




## Ready for broker use

Within ±20%: alley houses 64%, apartments 60%. Street-front houses 42% and land 43% need more work.

## Always show the 90% price range

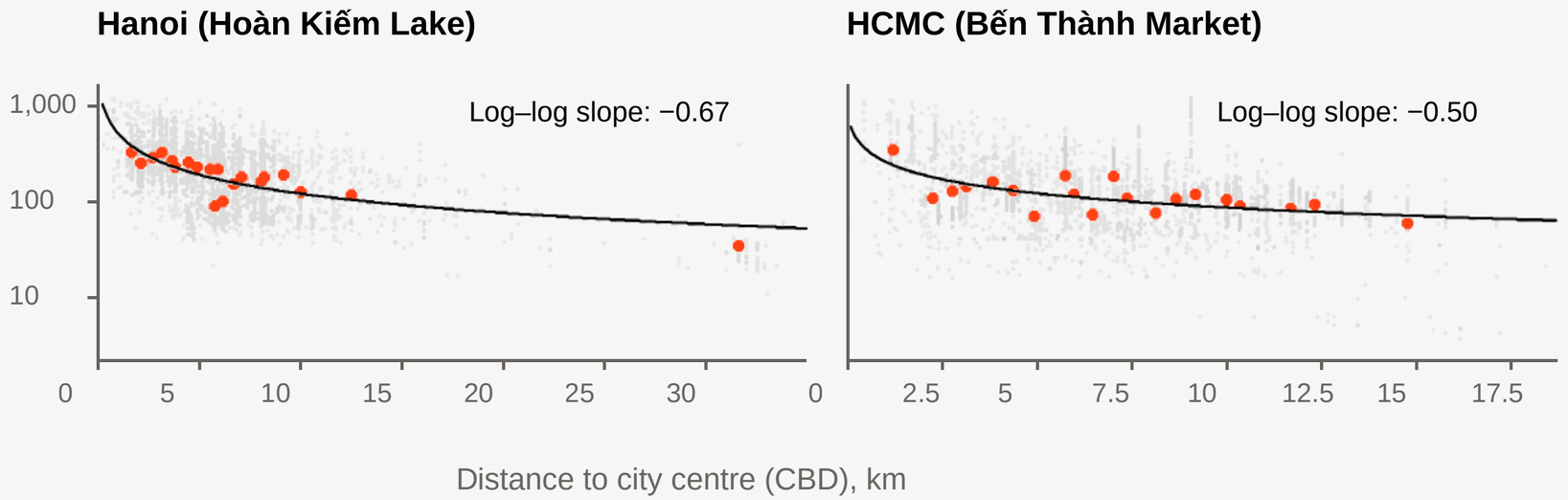
Half of estimates miss by over 18.1%. Log-price error is 0.354; the target is 0.20.

Source: mogi.vn listings, 15 Oct 2026 (N = 5,268); 1,054 test homes unseen in training. PPE20 bar rounded to 54%.

Map data gives 60% of the accuracy gain from district data.

# MAP DATA

Asking price (million VND/m²; log scale)




Points and trend lines redrawn from the source chart

## R² 0.714

Fit: size, rooms and legal status

**Adding district cuts error by 25%.**

**Map distances alone cut error by 15%.**

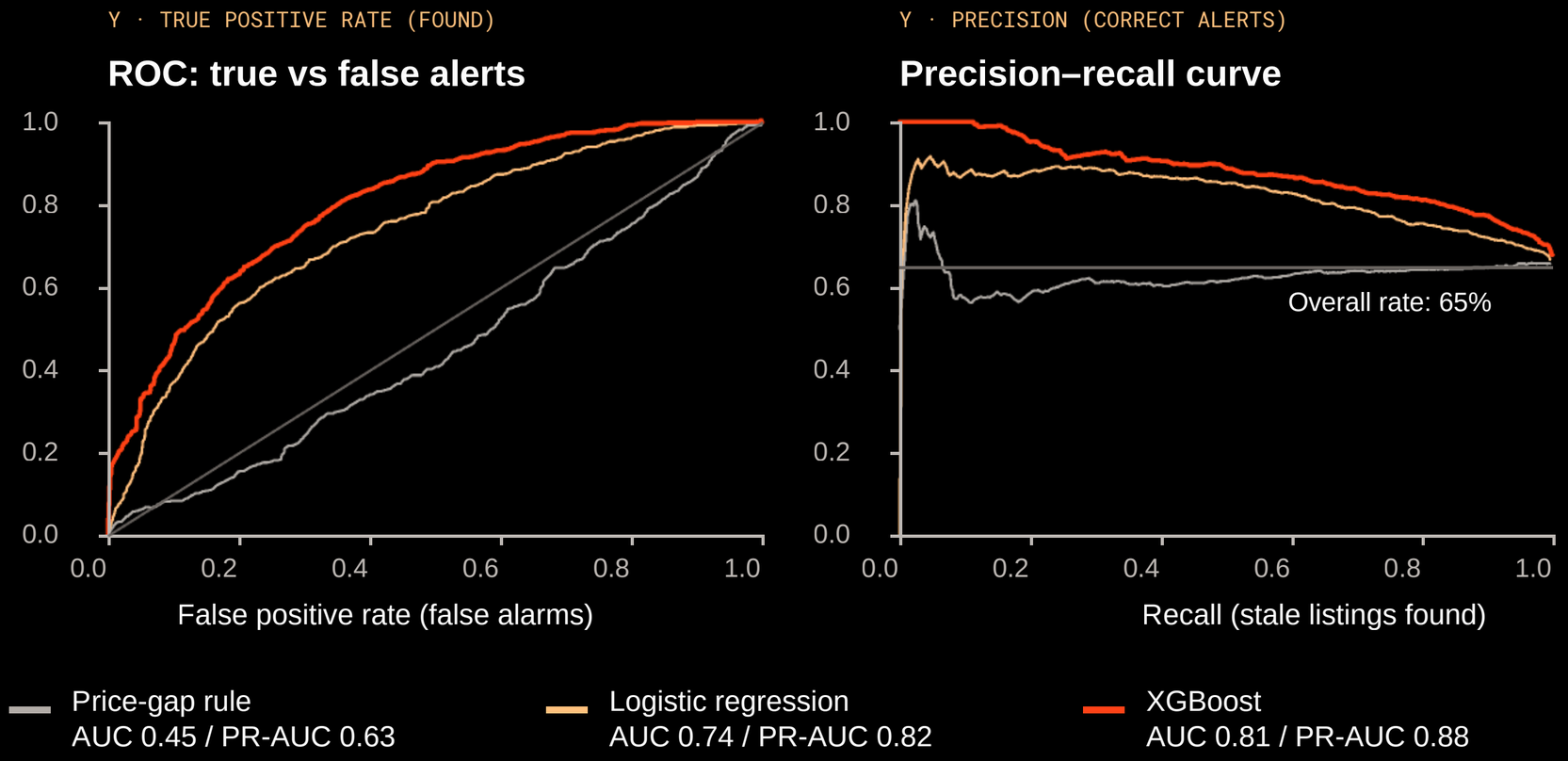
Values homes in districts with few listings.

## Map data helps value homes where listings are scarce

Source: mogi.vn, 15 Oct 2026 (N = 5,268); OpenStreetMap (ODbL). Regression uses log asking prices; error checked on held-out folds (cross-validation).

# STALE LISTING RADAR

Top 10%: 99% unsold over 180 days; 65% across all test listings.




## AUC 0.81

AUC = ranking score; 0.5 = guessing

Price-gap rule: no better than guessing. Price gap ranks 12 of 26 risk factors.

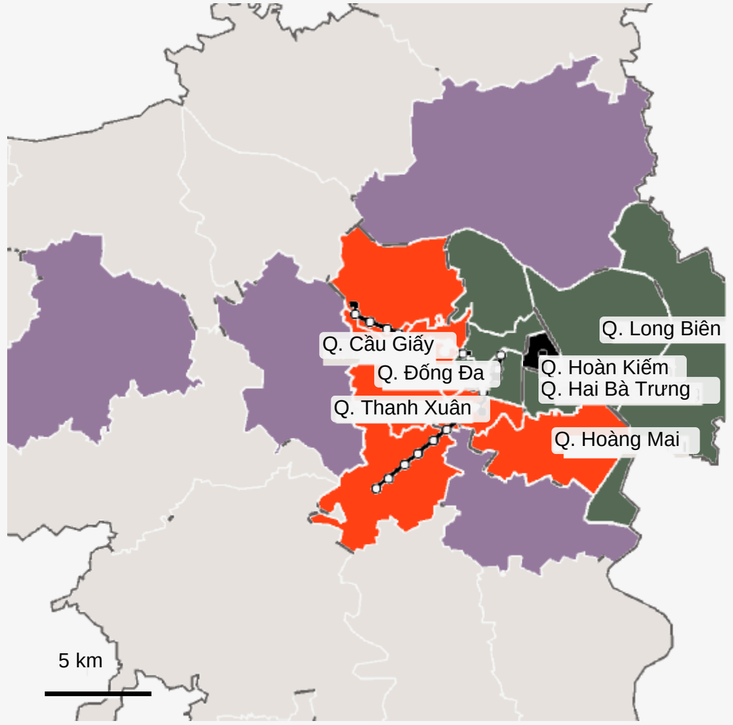
**Use radar to focus work. Use price gap to advise sellers.**

Source: mogi.vn listings, 15 Oct 2026 (N = 5,268); 1,054 test listings. Curves redrawn from the published chart.

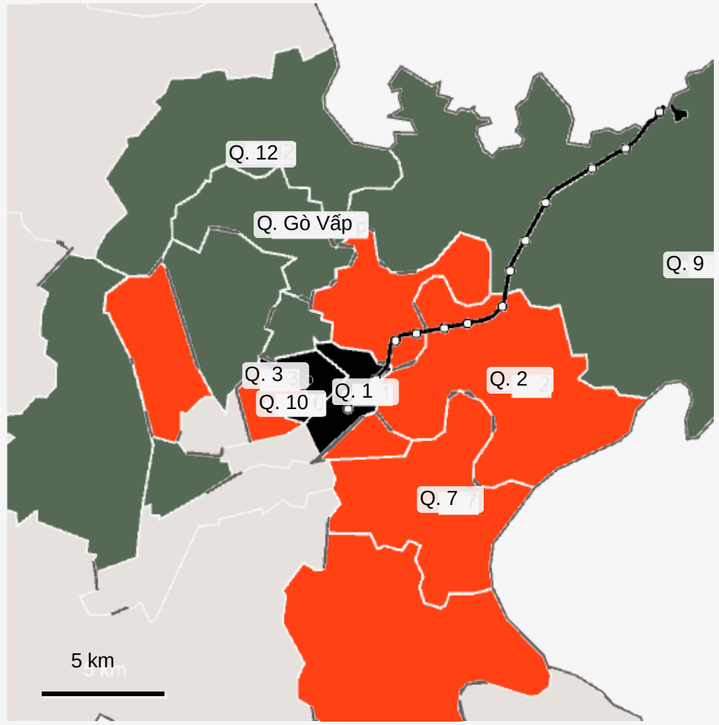
Four district groups need four different messages.

# MARKET MAP

**Hanoi**




**Ho Chi Minh City**




Prime core
Peripheral growth
Apartment mid-market
Too few listings (< 30)
Townhouse mid-market
Active metro line and station

## 01 Prime core

415 million VND/m². Hold price.

## 02 Apartment mid-market

Slowest: 364 days. Increase sales effort.

## 03 Townhouse mid-market

Fastest: 254 days. Sell more listings.

## 04 Peripheral growth

Mostly land. Highest stale risk: 36%.

Groups overlap. Use them for targeting, not fixed borders.

Source: mogi.vn, 15 Oct 2026 (N = 5,268); OpenStreetMap contributors (ODbL); district boundaries GADM 4.1.

# TOOLS

## SELECTION CRITERIA

- **Scalability**
- **Collaboration & Reproducibility**
- **Manager Usability**
- **Cost Efficiency**
- **Explainability**
- **Data Privacy & Access**

| LAYER | TOOLS | OUTPUTS | WHY IT FITS |
|---|---|---|---|
| 01 **DATA COLLECTION** | Python (requests, BeautifulSoup)<br>OpenStreetMap / Overpass · GADM | Property listings, amenities and administrative boundaries | **Scale & cost:** 7,792 listings/hour; lower paid map API costs. |
| 02 **STORAGE** | CSV · JSON · Parquet<br>Planned: cloud spatial database | Integrated dataset of 5,268 homes | **Scalability:** local prototype; planned cloud for shared access. |
| 03 **PROCESSING & MODELING** | pandas · GeoPandas · scikit-learn<br>statsmodels · XGBoost · SHAP | Valuations + 90% intervals<br>Stale-listing risk · Segments | **Explainability:** regression baseline; SHAP explains XGBoost estimates. |
| 04 **PRESENTATION** | Streamlit · HTML/JavaScript<br>matplotlib · Folium | Valuation dashboard<br>Market and district summaries | **Usability:** no-code access; HTML matched Python on 500 listings. |
| 05 **OPERATIONS & GOVERNANCE** | run_all.py · pytest<br>Planned: scheduling & monitoring | Reproducible pipeline<br>Planned model oversight | **Reproducibility:** automated tests; planned model-change approvals. |

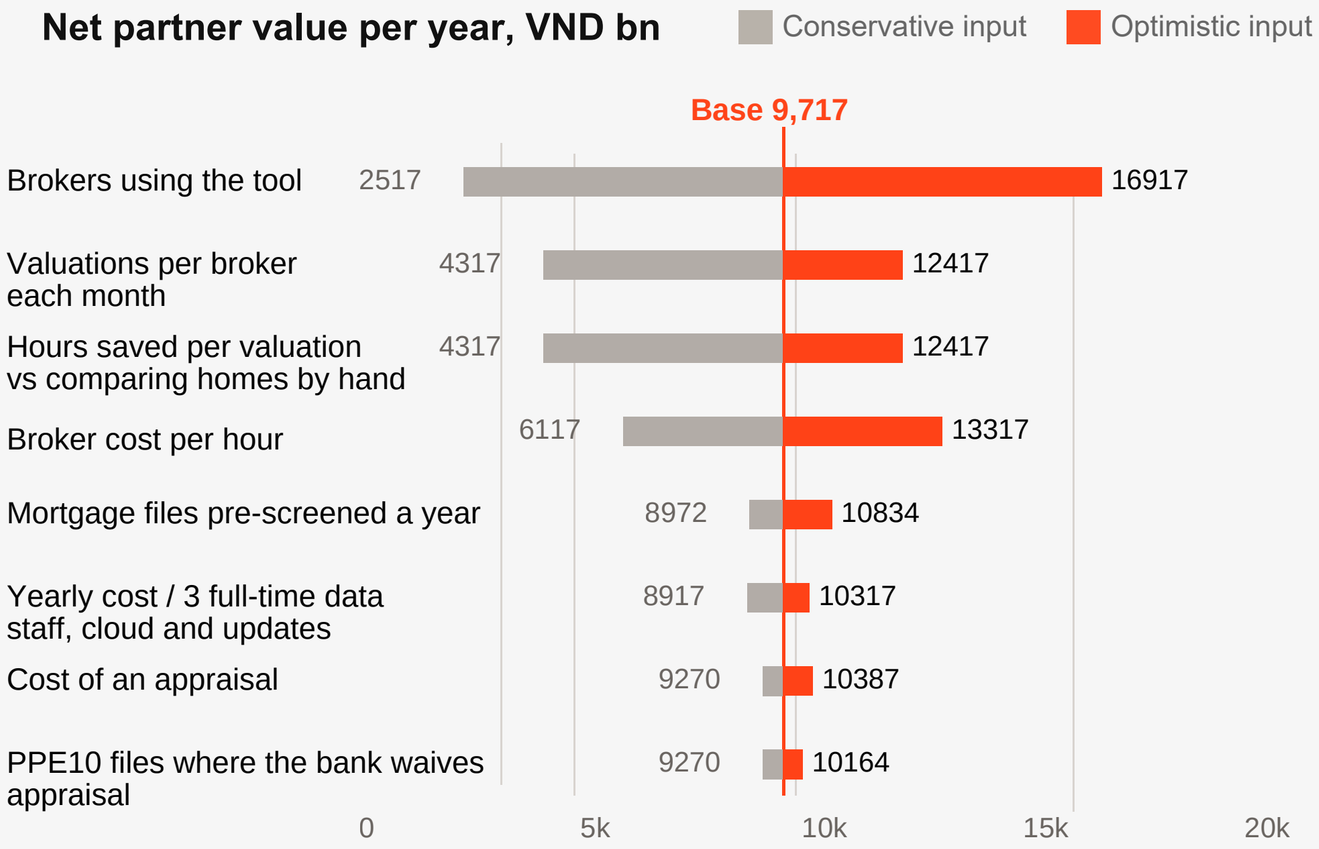
Each team gets a decision tool and a performance measure.

# DECISION TOOLS

| Team | Decision | Evidence | Measure (KPI) |
|---|---|---|---|
| **Sales and brokerage** | Which listings need attention, and which sellers need advice | Top 25% of risk scores: 70% of Quận 10 listings (HCMC). 37% of street-front houses ask >15% above fair value. | Days to sale: advised sellers vs other sellers |
| **Marketing and research** | Which areas to target, and what to say | Four district groups differ in price, selling time and home types | Buyer enquiries per listing, by group |
| **Risk and credit** | When to use a value estimate for an initial loan check | For 25% of homes, fair value is >20% above asking price. A desk review is needed. | Initial checks confirmed within ±10% |
| **Executive management** | Where to add brokers and offices | Selling speed vs price, by district | Median listing age, by group |

Adoption drives value: VND 9.7 bn net a year in the base case, 4.4× run cost.

# VALUE AND ROI




**9.7**

**VND bn net partner value a year**

## = 4.4× the run cost

Partner value 11.9 (broker time 10.8 + bank appraisal 1.1) − run cost 2.2

- Range, all inputs conservative → all optimistic: VND −2.3 to 40.9 bn
- Break-even at base inputs: about 30 active brokers with the bank saving, about 60 without
- Broker time is 91% of partner value
- Buyers avoid about VND 10.4 bn of overpayment (scenario, outside the ROI)
- Steady state at 300 active brokers; year one is lower

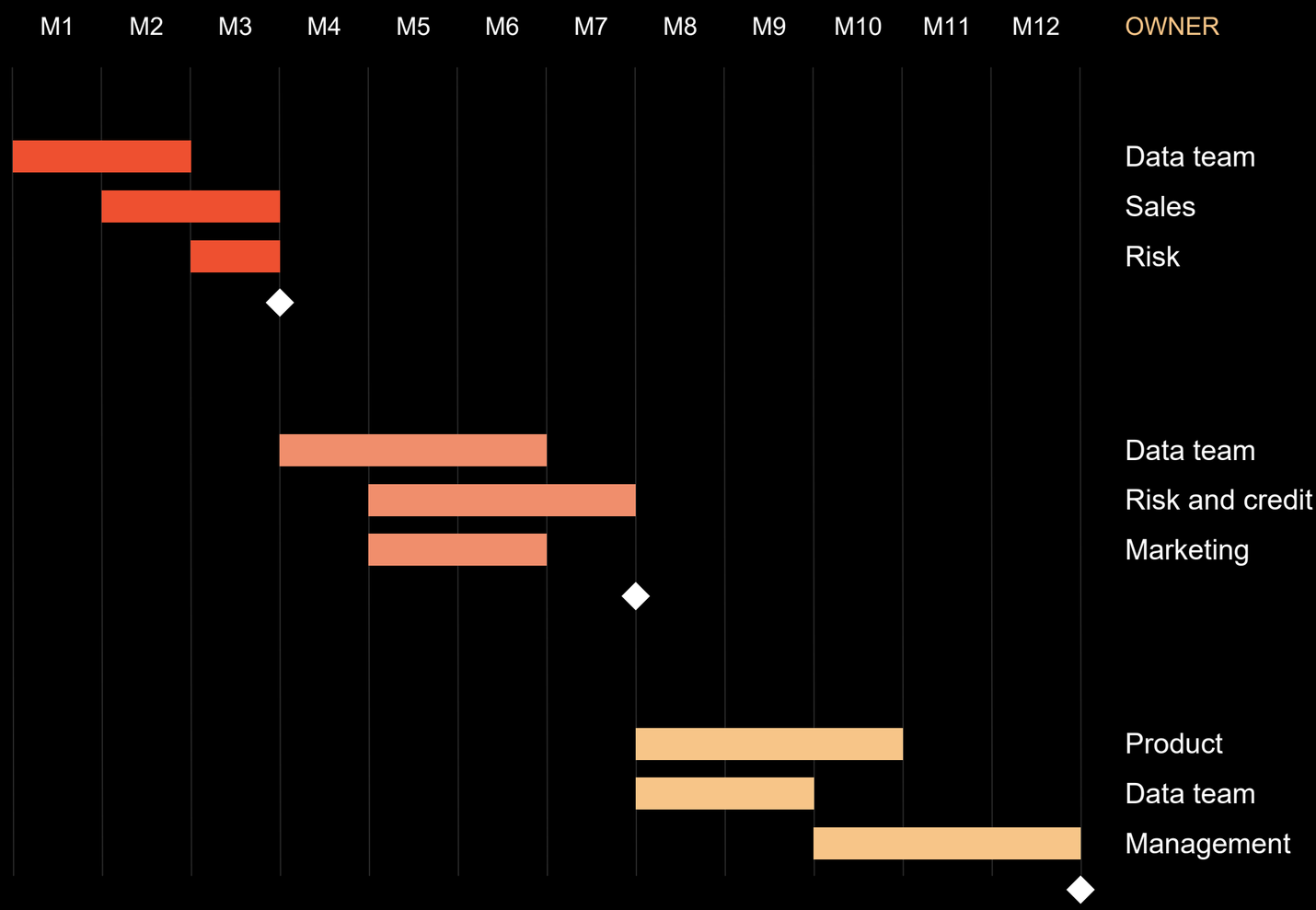
**Remember: value depends on brokers using the tool, not on model accuracy. We plan for 300 active brokers; break-even is about 60 at base usage.**



Scenario estimates on stated assumptions, not measured results. The 4.4× is an estimated benefit-to-cost ratio: broker time savings are monetised productivity, not necessarily cash. Valuations per broker and hours saved have the same effect because both multiply broker time saved.

12 months, three phases, and a continue-or-stop decision at each step.

# IMPLEMENTATION ROADMAP




## 01 PILOT / M1–M3

| | |
|---|---|
| Weekly data collection and database | Data team |
| Broker pilot in one agency | Sales |
| Monthly accuracy report | Risk |

**Gate 1 (M3): 60%+ within ±20% on pilot deals; weekly broker use**

## 02 SCALE / M4–M7

| | |
|---|---|
| Second portal and transaction prices | Data team |
| Bank pre-screening trial | Risk and credit |
| Segment reports for campaigns | Marketing |

**Gate 2 (M7): Bank waiver rule agreed; 300 active brokers**

## 03 PRODUCTISE / M8–M12

| | |
|---|---|
| Public valuation page and API | Product |
| Model version tracking, drift alerts | Data team |
| Quarterly market index for executives | Management |

**Gate 3 (M12): Accuracy published, stable 3 months; legal review done**

**Scale only when the numbers say so: accuracy and usage, then the bank agreement and 300 brokers, then published accuracy and legal clearance.**

Six known risks and three requests for a 3-month pilot.

# RISKS AND PILOT ASK

MAIN RISKS + RESPONSES

**Asking prices vs sale prices** — H / H
Calibrate on bank appraisals in Phase 2.

**Brokers adopt too slowly** — M / H
Track active brokers monthly; Gate 2 needs 300.

**Noisy unsold-listing label** — H / M
Weekly collection gives true time to sale.

**Website changes or blocks** — M / H
Tested page readers; second portal.

**Accuracy drifts as prices move** — M / M
Monthly report; re-train if PPE20 drops 5 points.

**Publishing home values** — L / H
Ranges, disclaimers, published method.

TO START THE PILOT

**01** One brokerage for a 3-month pilot (apartments, alley houses). Track active brokers, valuations per broker and hours saved.

**02** Appraised or notarised sale prices from a partner bank.

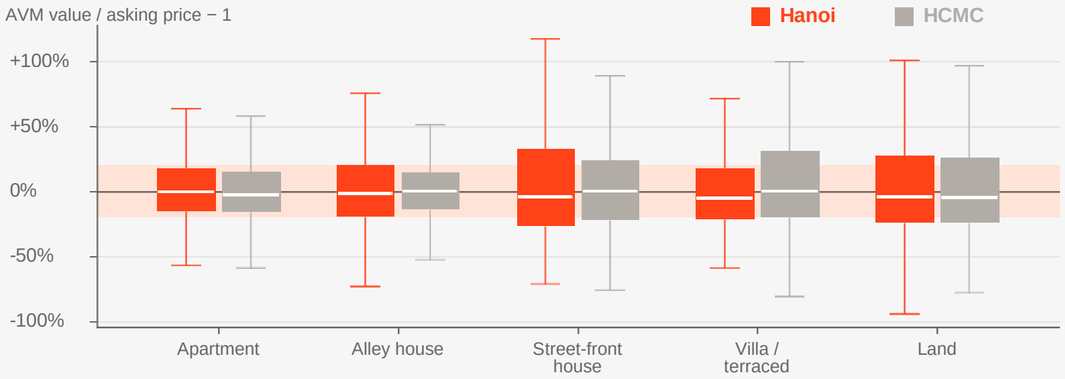
**03** Fund a 3-person data team: VND 2.2 bn a year in the base case.

## 3-MONTH PILOT

# APPENDIX

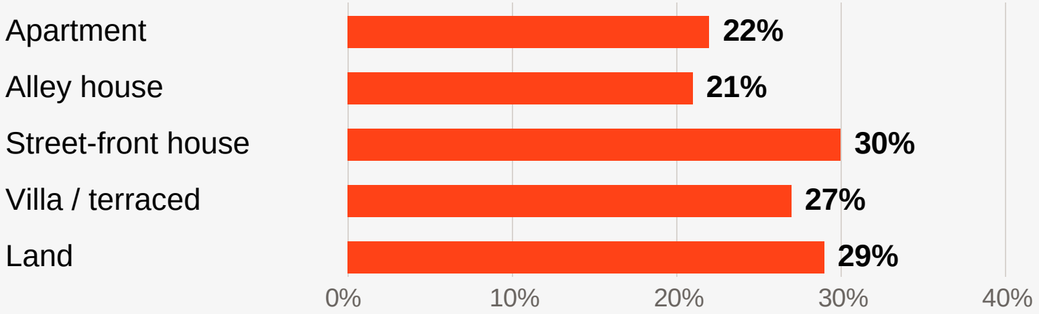
Expensive homes tend to be undervalued. Check this before any lending use.

## Errors on homes not used for training




Box: middle 50% (P25–P75). Shading: ±20%.

## Loan-security overvaluation risk




Share of homes with AVM value >20% above asking price

Source: mogi.vn listings, 15 Oct 2026 (N = 5,268); dashboard prototype.

## Vietnam Real Estate AVM

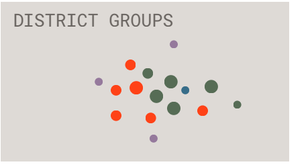
Prototype trained on mogi.vn asking prices (Oct 2026) and OpenStreetMap data. A guide, not a formal appraisal.

| Estimated fair value | 90% value range | Value per m² |
|---|---|---|
| **10.89 bn** | **6.85–17.34** | **210 m** |
| VND | bn VND | VND / m² |

Location data uses the ward median (middle value) of listed homes.

### District group

Townhouse mid-market: median 269 million VND/m², median age 259 days, 238 listings.




### Check the asking price

| Asking vs fair value | Stale risk (>180 days) |
|---|---|
| **−12.8%** | **Low (75%)** |

Ranks listing age; not the chance of a sale.

### Property

| | |
|---|---|
| City | Hà Nội |
| District | Quận Đống Đa |
| Ward | Select a ward |
| Type | Alley house |
| Area | 52.00 m² |
| Bedrooms | 4 |

### Similar listings

| District | Ward | m² | Beds | VND bn | m VND/m² | Deed | Listed |
|---|---|---|---|---|---|---|---|
| Quận Đống Đa | Xã Đàn | 50.6 | 4 | 15.5 | 306.3241 | Sổ đỏ | 15/10/2026 |
| Quận Đống Đa | Phường Văn Chương | 52 | 4 | 21 | 403.8462 | Sổ đỏ | 28/05/2026 |
| Quận Đống Đa | Phường Khâm Thiên | 48 | 4 | 5.5 | 114.5833 | Sổ đỏ | 17/10/2024 |
| Quận Đống Đa | Xã Đàn | 45 | 4 | 7.35 | 163.3333 | Sổ đỏ | 16/10/2024 |
| Quận Đống Đa | Phường Ô Chợ Dừa | 49 | 3 | 9.3 | 189.7959 | Sổ đỏ | 28/05/2026 |

Value, range, price gap, unsold risk, similar homes.

# CONTRIBUTIONS

| FULL NAME | STUDENT ID | KEY RESPONSIBILITIES | CONTRIBUTION |
|---|---|---|---|
| **Đinh Thị Ngọc Anh** | 2313560003 | Stale-listing model; Tools section; Presentation | **12.5%** |
| **Vũ Ngọc Ánh** | 2313560005 | EDA & cleaning; Stale-listing model; Communication & Strategy; Presentation | **12.5%** |
| **Đào Khánh Chi** | 2313560008 | Data section; EDA & cleaning; Fair-value model (XGBoost); Report editing & formatting | **12.5%** |
| **Nguyễn Thảo Linh** | 2312560022 | Fair-value model (OLS); HTML dashboard; Communication & Strategy; Presentation | **12.5%** |
| **Vũ Hoàng Phong** | 2312560034 | Problem statement; Stale-listing model; Prototype & slide editing; Conclusion | **12.5%** |
| **Nguyễn Hà Phương** | 2313560036 | Problem statement; Executive summary & report format; Fair-value model (XGBoost); HTML dashboard | **12.5%** |
| **Trần Minh Huyền** | 2213560025 | Data crawling; Method section; Market Map (K-Means) | **12.5%** |
| **Hà Lê Hoàng** | 2212560020 | Data crawling; Method section; Market Map; Final report check | **12.5%** |

**TOTAL TEAM CONTRIBUTION** **100%**

GROUP 1

# THANK YOU